

Using Moderator Analysis to Address Skepticism of Statistical Meta-Analyses in Education Research

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Abstract

Meta-analysis is a powerful statistical method for synthesizing analytic results from disparate studies. However, meta-analytic methods assume the comparability of studies, an assumption which is often violated or overlooked. This shortcoming can have extreme negative consequences on the validity of conclusions and therefore decision-making in the real world. This report argues that incomparability concerns can be significantly reduced by completing thorough moderator analyses within a rigorous analytic framework. These ideas are demonstrated via case studies on sub-group analysis, multilevel modeling, and meta-regression, all of which within the context of education research. An introduction to meta-analysis in general and a discussion on the limitations of the framework are provided.

1 Introduction

1.1 Background

Statistical meta-analysis has played a critical and unmistakable role in numerous fields since its creation in the mid-1970s, and comes with several advantages bolstered by practical utility. Hartung, Knapp, and Sinha (2008) say the intention of meta-analysis is to “provide the statistical analysis of a large collection of analysis results from individual studies for the purpose of integrating these findings”. In other words, the primary goal is to amplify effects analyzed in smaller studies which may have weaker implications on their own. A natural by-product is the potential to reconcile discrepancies in results between studies, benefits that are particularly relevant to the social sciences of education, sociology, and psychology. However, effect size amplification is compromised if the small studies are not the same or *comparable* in the first place. These fields are frequently challenged by differences in study design and the existence of confounding variables which lead to contradictory results, even if the studies examined identical research questions. Small studies in particular can vary greatly in terms of the covariates measured, measurement protocols, subject populations, and data quality checks, to name a few. In education research, common sources of variability are teacher quality, student motivation, demographics, and study quality itself. Other confounding factors may be quite subtle, making them difficult or impossible to obtain data for. Obtaining a sufficiently *comparable* sample of studies is one of the greatest challenges in meta-analysis, arising before any statistical methods are applied.

The assumption of comparability is often overlooked, which could ultimately invalidate analytical results. This paper briefly explores criticisms and the effect of poorly formulated meta-analyses in educational studies. A framework is then proposed which summarizes effective procedures for completing high-quality meta-analyses using moderator analysis. Next, techniques for detecting study-incomparability (in the form of heterogeneity and moderators) are covered. Upon detection, the methods of sub-group analysis, multilevel modeling, and meta-regression are motivated. Each method is demonstrated through a respective case study with real educational applications. Strengths and weaknesses are also provided for each method. The case studies are then related back to the framework via an assessment on the handling of study comparability concerns. Although a direct solution to the comparability problem may not be available, the methods are at least effective in controlling for it and potentially informative of its causes.

1.2 Criticisms and Negative Consequences

Like many statistical techniques, meta-analysis is often criticized for various shortcomings. Hansford and Schecter (2023) list the following common criticisms:

1. *“Apples-to-oranges” comparisons*: the comparisons of studies which are not fundamentally the same. For example, studies that seek the mean effect of comprehension instruction are fundamentally different from studies that analyze multiple types of comprehension simultaneously. Vocabulary and strategy are both used to teach comprehension, but are very different approaches. Studies that vary in this way may not be comparable under basic meta-analytic approaches.
2. *Ignorance of study quality*: many meta-analyses do not adequately account for differences in sample size, duration, attrition, or measurement methods between

studies. All of these are major contributors to study quality, and higher-quality studies are known to show lower pooled effects on average. Typically, randomized controlled trials (RCTs) are of a higher quality than quasi-experimental studies, which are, in turn, of a higher quality than simple case studies.

3. *Random, non-meaningful results:* meta-analysis captures random effects rather than trends. Issues may arise from the mathematical pooling (aggregation) of effect sizes. For example, taking the average effect size of a sample of 10 studies may have less meaning than the original sample due to variability compression. Analysts must ensure that information is not lost by combining effect sizes.

The negative effects of a study that does not account for these shortcomings can be catastrophic. The trivial outcome is the invalidation of the findings entirely should the critiques be ignored as it is illogical to draw an overarching conclusion about an effect size for studies that are incomparable. Incomparability also calls into question the fundamental principle of replication — how can results be trustworthy if they are not replicable? How can an analysis be replicable if the studies it uses are not comparable?

An example of a poorly-formulated meta-analysis is on exploring the effectiveness of two teaching methods, with a typical Cohen's d effect size to measure standardized mean differences in student achievement. Teacher quality is an obvious confounding factor, as one expects studies with higher-quality teachers to exhibit systematically different effect sizes than those with lower-quality teachers. Additionally, certain teaching methods may be more suited for particular teachers, students, or courses, creating more variability and perhaps constituting fundamentally different studies. Issues arise when the study is not designed to consider these confounding factors because its focus is on the teaching method. Other confounding factors are demographics and geography — is it reasonable to conclude that the effectiveness of a new teaching method is the same for students in rural Mississippi as for students in Manhattan? Meta-analyses must consider these study design features and incorporate them statistically wherever possible to ensure comparability and validate results. Failing to do so could ultimately lead to misallocation of time, money, and other resources by investing in spurious analytical findings. Poorly-executed meta-analyses can have legitimate consequences on the real world and peoples' lives.

1.3 Additional Notes

A key consideration of meta-analyses in general is the distinction between fixed effect and random effects models. The fixed effect approach assumes one true effect size between all studies, with variance among study-level effects deriving from sampling error alone. However, if the population variance is not small or close to zero, a random effects approach may be used. This approach assumes a *distribution* of effect sizes, from which analyses will estimate the mean effect and heterogeneity (between-study variability). In this setting, the variability among study-level effect sizes comes from two sources — sampling error *and* heterogeneity. The choice of model is particularly relevant to effective meta-analyses as using the incorrect form can lead to fundamental misunderstandings and faulty conclusions.

Another potential inhibitor to an effective meta-analysis is publication bias, which is beyond the scope of this paper. It is also challenging to adequately assess the effect of such bias on an entire field of study without access to a complete research registry. However, the possibility of bias or outlier studies in meta-analyses must always be addressed because

moderator analysis methods assume that missing (unpublished) studies would not have a significant effect on results.

Well-formulated meta-analyses address all potential shortcomings, thereby circumventing negative consequences. Moderator analysis works in various ways to achieve this, primarily by accounting for study comparability and quality differences. Section 2 details an effective meta-analytic framework with topics elaborated in Section 3 and demonstrated in Section 4. Section 5 contains closing thoughts and critiques of the framework.

2 Framework: Assessment of Comparability

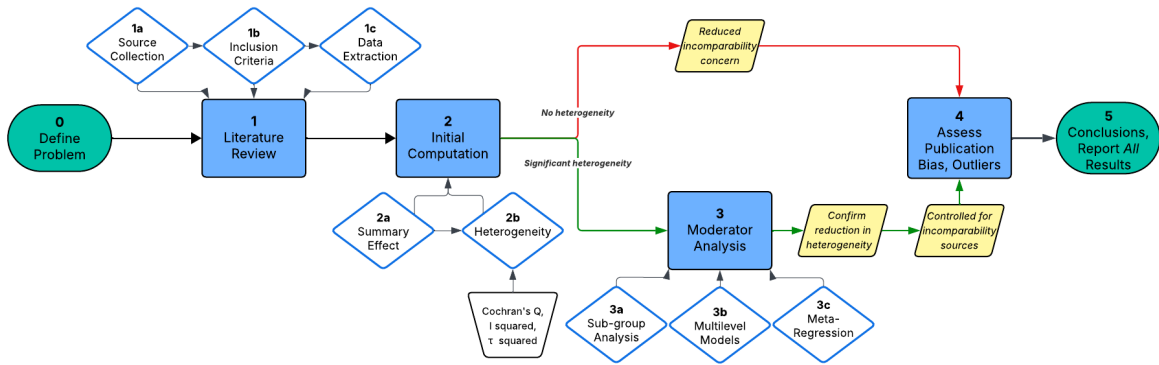


Figure 1: Assessment of Comparability Framework

Figure 1 outlines the suggested framework through which meta-analyses can be conducted to reduce comparability concerns as much as possible. Note that some important sub-calculations and details are omitted for brevity. These will be mentioned in the following explanation of the chart and the subsequent case study demonstrations.

Of course, the preliminary step of any analytical procedure is to establish a concrete research objective, denoted by Step 0. The natural follow-up is to perform a literature review, involving the collection of sources, the establishment of inclusion criteria, and the extraction of data. A quality meta-analysis is *highly* dependent on the literature review preceding it. Since the primary objective is to synthesize multiple analyses, the synthesis is only as good as the components being synthesized. Researchers begin by collecting sources related to the research problem using libraries (online or otherwise) and databases, represented by Step 1a in the chart. The collection may involve primary studies, disparate meta-analyses, or both. This leads to the creation of strict study selection criteria, which is absolutely critical to address comparability concerns. Sources must be sufficiently relevant to the research topic, contain sufficient data (including covariates where applicable), and be of satisfactory quality and recency. Additional, more specific criteria may be required depending on context. Once sources have been identified, data can be gathered for the new analysis.

Step 2 involves the initial results computation, which influences the future action to be taken. The initial estimate of the summary effect is computed here, typically with inverse-variance weighting to account for discrepancies in the precision of primary studies. An equally important part of this step is the estimation of heterogeneity, which

reflects variability between studies. Cochran's Q , I^2 , and τ^2 are discussed in the next section as statistics revealing the existence of heterogeneity and thus moderating effects. If significant heterogeneity is detected, random effects models and moderator analysis should be implemented. If no heterogeneity is detected, the incomparability concern is reduced but should not be completely written off. Researchers may still opt to explore the potential effect of moderators to further reduce concerns and reveal additional patterns in the data.

Step 3 is the primary focus of this report. It involves the application of moderator analysis techniques with the objective of revealing causes of heterogeneity and thus incomparability. Sub-group analysis, multilevel models, and meta-regression serve as effective techniques to achieve these goals. Where applicable, it is best to complete as many moderator analysis techniques as possible, as permitted by data structure and availability. Various computational methods can be used within each method as well, providing an understanding of the robustness of results to a range of approaches. The techniques allow researchers to reduce heterogeneity between studies by controlling for potential sources of incomparability, reducing such concerns.

Step 4 assesses the effect of publication bias and outliers on meta-analytic results. Although not covered in great detail in this report, this step is necessary to address the additional shortcomings of the analysis. Outlier assessment allows for sensitivity analysis and may confirm the robustness of results, making them more reliable. A meta-analysis may be considered incomplete if it does not address the possibility of publication bias and report the related analytical results.

Lastly, once all procedures have been completed, researchers draw analytical conclusions and report the results of *each* step. If no steps are skipped, researchers assure the audience and future analysts of the replicability, reliability, and robustness of their findings. Meta-analysis is inherently an ongoing process, and no report is absolutely *final* in its conclusions. There will always be value in completing additional studies, collecting more data, and synthesizing more results.

3 Moderator Analysis Overview

Moderator analysis is a broad term referring to methods that seek to identify moderators, or predictors, that explain variability in the effect sizes in a meta-analysis. It essentially seeks to identify potential confounding factors such as those discussed in Section 1. Moderator analysis can be viewed as an umbrella of techniques that elucidate study comparability by *identifying factors that create doubt in the synthesis of their results*. Moderators can be subdivided into substantive or methodological characteristics, the former dealing with study objectives and the latter relating to study design. These align with the first and second critiques of Section 1.2 and motivate the use of moderator analysis as a means of addressing study incomparability. If the existence of significant moderators is detected, the analyst may have increased concerns, but additional procedures must be completed to confirm the existence of moderators and determine whether they actually have an important effect on results. Once moderators are controlled for and analysis is completed, comparability concerns may be reduced since the *cause* of discrepancies will have been identified and the variance in effect sizes will have been reduced by the inclusion of additional predictors. Moderator analysis improves the analyst's understanding of the phenomena they are researching while addressing each of the primary

meta-analysis’s shortcomings. Although correlations between characteristics and effect sizes do not necessarily imply that the characteristics are *causing* comparability concerns, testing for moderators is imperative to sufficiently understand and validate meta-analytic results, regardless of whether or not there are legitimate concerns of incomparability.

3.1 Detection Methods

The initial computation of a meta-analysis should include the calculation of heterogeneity statistics which may indicate the existence of moderating effects. Three statistics reflecting the level of heterogeneity are listed below.

- *Cochran’s Q and χ^2 test*: This method tests for the homogeneity, or constant variance, of true effect sizes across all studies. It is commonly used despite its low statistical power and sensitivity to sample size (larger samples of studies increase the likelihood of detection). Q is calculated as

$$Q = \sum_{k=1}^K w_k(\theta_k - \theta)^2 \quad (1)$$

where K is the total number of studies, and w_k and θ_k are the inverse-variance (precision) and effect size of study k , respectively. θ is the fixed-effect pooled estimate for the overall effect size, a sample mean weighted by the precision of each study. Inverse-variance weighting is utilized so that more precise (larger) studies have a larger effect on results than less precise (smaller) studies. Q follows a chi-square distribution, allowing for significance tests. A significant result and thus rejection of the null hypothesis may allow the researcher to conclude that there is significant heterogeneity (and thus some moderating effect) among the studies.

- I^2 : This statistic represents the percentage of variability in effect sizes not caused by random error and instead caused by heterogeneity. It is calculated as

$$I^2 = \frac{Q - (K - 1)}{Q} \quad (2)$$

with K as the number of studies and Q calculated as above. It is often interpreted by quartiles, representing thresholds for low, medium, and high heterogeneity. Larger values therefore carry greater suspicion of moderators.

- τ^2 : This statistic estimates the amount of variance in true effect sizes among the studies in the meta-analytic sample. Its magnitude directly reflects the degree of heterogeneity, and is usually calculated using DerSimonian-Laird’s method:

$$\tau^2 = \frac{Q - (K - 1)}{\sum w_k - \frac{\sum w_k^2}{\sum w_k}} \quad (3)$$

K , Q , and w_k are the same as above. Larger values signify greater heterogeneity and perhaps the existence of moderator variables.

If significant heterogeneity is detected, a random effects meta-analysis should be applied since the population variance is non-zero. Detection also increases suspicion of moderators and, therefore, study incomparability, meaning additional methods should be applied to confirm and understand the effect of said moderators. These methods enable reliable analyses and the effective handling of comparability concerns.

4 Case Studies

This section examines three studies in education research that used each of the selected methods within moderator analysis. Explanations of each method are provided, followed by respective case studies. The background, methods, and results of each study will be summarized along with a discussion through the lens of the framework established in Section 3, providing proofs of concept and connecting the research back to study comparability concerns.

4.1 Sub-Group Analysis

4.1.1 Method Description

Sub-group analysis is a straightforward method of moderator analysis and thus of revealing study comparability issues. For this approach, studies are grouped based on a categorical variable and statistics are compared across the resultant groups. It is equivalent to weighted ANOVA and almost always assumes a random effects model. Comparisons are made between effect size point estimates, standard errors, confidence intervals, and heterogeneity statistics (Q , I^2 , τ^2) for each group. In the context of subgroup analysis, τ^2 reflects the dispersion of true effects within a sub-group, and is thus calculated within each group. Higher values of τ^2 of course signify greater discrepancies between groups, implying a moderating effect and perhaps increasing study comparability concerns. I^2 is often calculated to estimate the proportion of variance explained by the moderators, and pairwise significance tests are used to compare effect size point estimates.

This approach is a natural response to the second critique in Section 1.2 about the ignorance of study quality. If the analyst knows the types of studies in the sample, they can easily divide the sample according to study design facets, and determine if there are any considerable differences in effect sizes between study type. When applicable, analyses may group randomized controlled trials, quasi-experimental studies, and case studies separately. Many other groupings are possible, so additional moderators may also be considered as data availability permits. Although sub-group analysis is limited to categorical variables, it is a quick and convenient way of establishing a specific moderating effect. Another downside is the potential for small sample sizes in sub-groups, resulting in low statistical power for hypothesis testing, a factor that must be considered before applying a given sub-grouping.

4.1.2 Paper Introduction and Background

The Power of Feedback Revisited: Meta-Analysis of Educational Feedback Research (2020) from Wisniewski, Zierer, and Hattie demonstrates the utility of sub-group analysis in education research. The effect of feedback on student learning was examined, a topic known for volatility in results. Feedback is one of the most important aspects in education because it encapsulates the fundamentals of teaching and helps to determine the optimal methods for fostering development. The authors asked the following: 1) What is the overall effect of feedback on student learning? and 2) To what extent is the effect of feedback moderated by specific feedback characteristics? Moderator analysis was motivated in their literature review, citing the importance of interpreting different forms of feedback as independent measures. They also advocate for random effects models, as there exists much theoretical background and prior evidence against homogeneity

because of moderating effects. They acknowledged questions on the production of interpretable results due to incomparability concerns arising from moderators, but argued that appropriately weighing each study and applying sub-group analysis to control for differing feedback characteristics addresses shortcomings sufficiently.

4.1.3 Implementation and Results

The authors used rigorous criteria for study selection, removing duplicates (resulting from the combination of multiple meta-analyses), studies with insufficient data reporting, and studies outside of the educational context. Dependent variables related to student learning, such as achievement, motivation, or behavioral change. The authors used Cohen’s d effect size for standardized mean difference between treatment groups with inverse-variance weighting under a random effects model, and controlled for moderating effects via sub-group analysis. They coded in six new variables to define sub-groups:

- Research Design (studies with control groups vs pre-post-test)
- Publication Type (e.g. journal article, dissertation)
- Outcome Measure (cognitive, motivational, physical, behavioral)
- Feedback Type (reinforcement vs punishment, corrective, high-information)
- Feedback Channel (written, oral, assisted by video, audio, or computer)
- Feedback Direction (e.g. teacher to learner, learner to teacher)

To justify the use of sub-group analysis, homogeneity must be observed in the effect sizes *within* groups, meaning there is less left over variability once moderators are controlled for. Q tests are used to check for homogeneity, along with the calculation of I^2 . The average effect sizes should also differ significantly *between* resultant groups. Table 1 summarizes the results.

Table 1: Tests of heterogeneity between and within sub-groups. *Wisniewski et al., 2020*

Moderator	$Q_B(df)$	p	$Q_W(df)$	p	I^2
Research design	29.06(1)	< 0.0001	5639.74(955)	< 0.0001	83.4%
Publication type	6.15(1)	< 0.05	5699.39(957)	< 0.0001	83.4%
Outcome measure	14.12(3)	< 0.001	4380.69(750)	< 0.0001	83.0%
Feedback type	41.52(2)	< 0.0001	1541.06(316)	< 0.0001	80.9%
Feedback channel	5.12(2)	> 0.05	2218.20(337)	< 0.0001	85.2%
Feedback direction	9.35(2)	< 0.001	4895(852)	< 0.0001	81.9%

Q_B : heterogeneity between groups, Q_W : heterogeneity within groups

I^2 : total amount of heterogeneity

All moderators showed significant heterogeneity *between* groups except feedback channel. Each moderator had significant heterogeneity *within* their groups, meaning that although variability was reduced by using the moderators, there is still a significant amount remaining. This is evidenced by the difference in magnitude between Q_B and Q_W for each moderator. While any reduction in heterogeneity can reduce comparability concerns, it can be said that more effective groupings are needed.

For the overall effect estimate, 994 effect sizes were used across 435 studies, calculated from 61,000 total subjects. The weighted average effect of feedback was estimated as

0.55, with a 95% confidence interval of (0.48, 0.62). 17% of the effects were negative, and $I^2 = 86.47\%$, with 70% of scores in experimental groups exceeding the average score in control groups.

Table 2 summarizes the difference in effect sizes between moderator groups. Feedback channel is not included as it was the only moderator with insignificant heterogeneity between groups. The results show larger effects for pre-post design studies than controlled studies and larger effects for journal articles than dissertations. Feedback has a larger effect on learning for cognitive and physical outcomes than for motivational and behavioral outcomes, despite considerable variability remaining in the cognitive outcome group. Corrective and high-informative feedback (i.e. specifics on the task or process a student is using) outperformed reinforcement/punishment, and student-to-student was the most effective feedback direction.

Table 2: Tests of heterogeneity between and within sub-groups. *Wisniewski et al., 2020*

Moderator	k	d	C.I.	Q	I^2
Research design					
<i>Controlled study</i>	713	0.42	[0.37 – 0.46]	3321.86	78.6%
<i>Pre-post study</i>	244	0.63	[0.56 – 0.69]	2317.88	89.5%
Publication type					
<i>Journal article</i>	843	0.49	[0.45 – 0.53]	5176.67	83.7%
<i>Dissertation</i>	116	0.36	[0.25 – 0.46]	522.72	78.0%
Outcome measure					
<i>Cognitive</i>	597	0.51	[0.46 – 0.55]	3689.88	83.8%
<i>Motivational</i>	109	0.33	[0.23 – 0.42]	600.98	82.0%
<i>Physical</i>	19	0.63	[0.34 – 0.92]	36.65	50.9%
<i>Behavioral</i>	30	0.48	[−0.09 – 1.06]	0.28	50.0%
Feedback type					
<i>Reinforcement or punishment</i>	39	0.24	[0.06 – 0.43]	123.64	69.2%
<i>Corrective</i>	328	0.46	[0.39 – 0.55]	1260.41	81.2%
<i>High-information</i>	42	0.99	[0.82 – 1.15]	157.12	73.9%
Feedback direction					
<i>Teacher > student</i>	812	0.47	[0.43 – 0.51]	4510.40	82.0%
<i>Student > teacher</i>	27	0.35	[0.13 – 0.56]	52.92	50.9%
<i>Student > student</i>	16	0.85	[0.59 – 1.11]	132.28	88.7%

k : number of study effects, d : average effect size, $C.I.$: 95% confidence interval

Q : heterogeneity within subgroup, I^2 : total amount of heterogeneity within subgroup

4.1.4 Discussion

The paper serves as a great example of the framework in that it accounts for any possible meta-analytic shortcoming, including the issues of comparability. Although omitted in this report for brevity, they addressed the possibility of outliers and publication bias (including funnel plots and Egger’s regression test). Strict study selection criteria are used, which, as established, are absolutely critical to address comparability concerns. They also used robust variance estimation to account for non-independence of study effects and avoid complications arising from studies reporting multiple effect sizes. Additionally, the authors motivate sub-group analysis as an appropriate method to capture the effects of moderators contributing to heterogeneity between studies.

Wisniewski, Zierer, and Hattie also provide a discussion on where future research may be needed on the topic, in that several sub-groups had very small sample sizes. For example, additional research on motivational and behavioral outcome measures, reinforcement

feedback, and student-to-student feedback may be motivated. They note that the "pursuit of seeking the optimal moderators is the core business of future research", implying that much more work is needed to strengthen conclusions for this topic. However, the authors were effective in maximizing the information to which they had access while applying the most pertinent tools to ensure analytic reliability.

4.2 Multilevel Modeling

4.2.1 Method Description

The multilevel modeling approach is motivated by data that follow some hierarchical structure, particularly data with at least three levels. Meta-analysis analyzes differences between studies, which in turn analyze differences between individual observations, where each observation may possess numerous features. This creates a hierarchical structure of three levels with data at Level 1, nested within participants at Level 2, nested within studies at Level 3. Multilevel modeling takes advantage of this structure by formally modeling the sources of variability and the dependency between effect sizes at each level. It is particularly strong when there are multiple effect sizes coming from the same population. Additionally, multilevel modeling can accommodate continuous *and* categorical moderator variables, and several moderators simultaneously, making it more flexible than sub-group analysis.

A common experimental design (in education and otherwise) is the single-case experimental design (SCED). These experiments involve repeated measurements on an individual under varying circumstances. Since meta-analyses compare the results of several of these studies, a three-level hierarchical data structure arises which can be modeled as:

$$Y_{ijk} = (\beta_0 + v_{0k} + u_{0jk}) + (\beta_1 + v_{1k} + u_{1jk})X_{ijk} + e_{ijk} \quad (4)$$

where i represents the measurement occasion, j represents the case, and k represents the study. v_{0k} and v_{1k} are Level 3 random *study* effects that represent how the baseline and treatment effect for study k deviate from the overall baseline and treatment effect (β_0 and β_1), respectively. u_{0jk} and u_{1jk} are Level 2 random *case* effects that represent how the baseline and treatment effect for case j from study k deviate from the baseline and treatment effect of study k . The estimated parameters of the model are the fixed regression coefficients (β 's) and the variance-covariance parameters for all random effects. Researchers may constrain covariances between random effects at a given level to zero to reduce the number of parameters to estimate. Note that X_{ijk} is a dummy indicator variable in this case, but the model can be extended to include additional covariates (at various levels), adding more β parameters. Alternatively, the model can be collapsed into a two- or one-level model. This is often the case in studies without repeated measurements on the same unit of observation.

Restricted maximum likelihood estimation (REML) is the most common method for parameter estimation, as it reduces the risk of overfitting by restricting to a transformed data set independent of the fixed-effect regression coefficients and performs well in small-sample settings. Empirical Bayes estimates may also be utilized to give more precise results, although it does not provide a complete quantification of probabilistic uncertainty of the hyperparameters. Testing multiple estimation methods is recommended to ensure robustness of results and account for their unique strengths and weaknesses. For

interpretation, multilevel modeling is typically concerned with the variance parameters of random effects as they indicate how much the baseline or covariate effects vary within their respective level. In the context of study comparability and moderator analysis, a random effect with a large variance may be identified as a major cause of incomparability. However, formally modeling it and capturing the variability it causes reduces concerns.

Multilevel modeling has several important assumptions that should *always* be checked whenever it is applied to ensure the validity of implementation and interpretations. In addition to the following, sufficient sample sizes, notably within each cluster of studies, are critical to reduce the risk of biased parameter estimates.

- The relationship between the moderator(s) and the effect size is linear
- Moderators are not highly correlated with each other (no collinearity) and uncorrelated with random effects
- Residual variance is constant across the range of covariates (homoscedasticity)
- The residuals and random effects (at each level) are multivariate normally distributed with zero mean and unknown variance-covariance matrix. Several specifications of the structure of the variance-covariance matrix can be tested
- The sampling errors of the effect sizes are independent
- The clusters (different studies) are independent of each other

Multivariate multilevel modeling is exceptional for addressing comparability concerns in that several moderators can be examined at once, capturing their covariance and difference in relations to covariates. Its flexibility also allows for the modeling of disparate outcome variables between studies, which can enable researchers to relax their study selection criteria while increasing statistical power by creating larger samples. However, the method is limited in that it can be computationally expensive, in particular for extremely large data sets. The computation time grows quickly as additional random effects are included, and convergence warnings or singular fits may occur. In addition, interpretation and policy implementation based on results can be tricky due to the hierarchical complexity of certain data sets. It is easy to place or interpret variables at an incorrect level, making misspecification common. Misspecification could be catastrophic in education research as it typically increases the Type I error rate, which can lead to poor decision-making in the real world. However, the level of control and flexibility offered by multilevel modeling make it a worthwhile method for effective meta-analyses.

4.2.2 Paper Introduction and Background

Meta-Analysis Using Multilevel Models with an Application to the Study of Class Size Effects (2002) by Goldstein et al will serve as the case study for this method. It serves as an excellent reference for the application of multilevel modeling in general and as a demonstration of its implementation. The effect of class size on student performance has been scrutinized for many decades and has implications on school design in terms of architecture, budgeting, and staffing. The authors applied multilevel models to combine study-level data with individual-level data. The standardized mean difference between the scores of students in larger classes and the scores of students in smaller classes served as the primary effect size. To account for the differing methods for measuring student

achievement and the correlation between pre- and post-treatment scores, the authors used adjusted means and standard deviations.

4.2.3 Implementation and Results

Like the first case study, Goldstein et al used a strict criteria for study inclusion, requiring studies with: 1) an RCT or matched design (to include/compare smaller and larger classes), using school or study-level criteria, 2) achievement score outcomes, 3) longitudinal data with initial achievement compared to final achievement, 4) small classes having at least 15 students, large classes having at most 40. This resulted in a small sample size of only 9 studies, but the studies were fairly representative of various regions in the United States (e.g. California, Massachusetts, Texas, Ohio).

The authors applied two approaches; modeling at the aggregate level, and modeling that combined individual-level data with study-level data (hierarchical modeling). Brief introductions to their methods and findings are provided by the following.

1. Aggregate-Level

For this approach, the authors treated class size as a continuous variable and centered it at 15 to improve interpretability. Grade level was incorporated as a fixed effect. Their model notation differs slightly from the notation introduced previously:

$$\begin{aligned}
 y_{.jk} &= \alpha_{0jk} + \alpha_{1k}C_{jk} + \sum_l \beta_l G_{ljk} + e_{.jk} \\
 \alpha_{0jk} &= \alpha_0 + u_{0jk} + v_{0k}; \quad \alpha_{1k} = \alpha_1 + v_{1k} \\
 u_{0jk} &\sim N(0, \sigma_{u0}^2); \quad e_{.jk} \sim N(0, \sigma_e^2/n_{jk}) \\
 v_{0k} &\sim N(0, \sigma_{v0}^2); \quad v_{1k} \sim N(0, \sigma_{v1}^2); \quad cov(v_{0k}, v_{1k}) = \sigma_{v01}
 \end{aligned} \tag{5}$$

j and k index the grade and study, respectively. α_0 estimates the mean score for a class size of 15, u_{0jk} is the random departure of the j th grade mean from the k th study. β_l is the fixed effect for grade l , with G_{ljk} containing dummy variables for the grades. v_{0k} is the residual for study k . C_{jk} represents the class size, and α_1 estimates the overall effect of class size per additional student. v_{1k} estimates the random departure for the k th study from the overall class size effect. A few variations of this model are tested.

- *Model A - Class size as a random effect across studies*

This model fit the data the best, having the greatest log-likelihood. Each additional student in a class reduces reading scores by 0.02 standard deviation units. Grade level effects were not significant.

- *Model B - Class size as a fixed effect across studies*

Relatively poor model fit, indicating substantial evidence of heterogeneity in class size effect between studies. This was confirmed via likelihood ratio test. Grade level remained an insignificant moderator.

- *Model C - The same as Model B, adding a quadratic term for class size effect*

There is some quadratic effect of class size — class sizes between 15 and 30 show a decrease in reading performance, but an increase in performance in classes with more than 30 students. This is influenced by one particular study having very large classes.

The authors point out the limitations of this aggregate approach in that it cannot incorporate individual-level data and thus reduces modeling flexibility and control over variability.

2. Combination

The authors then shifted to analyze the STAR study, which had three levels (school, class, and student). In this study, students were assigned to classes of 13-17 or classes of 22-25 and followed from kindergarten entry through third grade. Reading test scores are once again the outcome variable of interest. Grade level was omitted in this modeling approach due to its insignificance at the aggregate level.

$$\begin{aligned}
y_{ijkl} &= (\alpha_{0l} + \alpha_{1l}C_{ijkl}) + e_{.jkl}(1 - z_1) + (\alpha_{0ijkl} + \alpha_{2l}x_{2ijkl} + v_{1kl}C_{ijkl})z_1 \\
\alpha_{0l} &= \alpha_0 + w_{0l}; \quad \alpha_{1l} = \alpha_1 + w_{1l} \\
\alpha_{0ijkl} &= v_{0kl} + u_{0jkl} + e_{ijkl} \\
z_1 &= 1 \text{ if individual-study data, } 0 \text{ otherwise} \\
w_{0l} &\sim N(0, \sigma_{w0}^2); \quad w_{1l} \sim N(0, \sigma_{w1}^2); \quad \text{cov}(w_{0l}, w_{1l}) = \sigma_{w01} \\
v_{0kl} &\sim N(0, \sigma_{v0}^2); \quad v_{1kl} \sim N(0, \sigma_{v1}^2); \quad \text{cov}(v_{0kl}, v_{1kl}) = \sigma_{v01} \\
u_{0jkl} &\sim N(0, \sigma_{u0}^2); \quad e_{ijkl} \sim N(0, \sigma_e^2); \quad e_{.jkl} \sim N(0, \sigma_e^2/n_{jkl});
\end{aligned} \tag{6}$$

The above constitutes a four-level model with students (*i*) grouped within classes (*j*) within schools (*k*) within studies (*l*). x_{2ijkl} represents the score at the end of kindergarten, while C still represents class size. σ_{w0}^2 is the *between-study* variance of class size effect, and σ_{v1}^2 is the *between-school* variance of class size effect.

Variability between studies was slightly smaller under this specification compared to the aggregate analysis. The quadratic effect of class size became negligible, and the linear effect of class size is less precise due to significant heterogeneity. In any case, the linear effect of class size was effectively the same as the aggregate analysis. There was small variation in class size effect between schools, though other studies have shown that math scores exhibit greater variability than reading scores. The authors noted that intra-class and intra-school level correlations are large, implying aggregate-level inferences may overestimate statistical significance.

4.2.4 Discussion

Despite its age, Goldstein et al.'s paper serves as a quality reference for moderator analysis application and consideration of study incomparability. Perhaps most importantly, it applies rigorous criteria for study inclusion, ensuring that they are as comparable and relevant as possible. A weakness of the paper is certainly the sample size, as they were only able to include 9 studies. Additionally, the authors acknowledged the concern of studies measuring the "same" thing (i.e. differing methods to determine student performance) and are thus aware of the issues of incomparability. However, it can be readily argued that the nature of multilevel modeling compensates for discrepancies in that it captures the inherent structure of the data, providing precise fitting methods. Calculating summary statistics such as heterogeneity, ICC, and variability in random coefficients also allows researchers to assess the level of comparability and establish avenues for future work. In any case, researchers can use intuition about the studies — as long as the whole picture is conveyed and models are fit appropriately, comparability concerns ought to be reduced.

The authors mention that the methods applied could be extended to multivariate multilevel modeling, which allows for several outcome variables. Although this brings about additional computational complexity, such an approach would eliminate the need for adjusted statistics. This could combine studies with math scores and reading scores measuring student achievement, for example. Thus, it allows for the interpretation of different relationship structures and correlations between variables. Researchers must avoid overfitting the data (by including too many study-level covariates), which would diminish reproducibility and external validation. Goldstein et al. also discuss the value of testing additional explanatory variables, motivated by similar objectives to Wisniewski, Zierer, and Hattie. They note that more work is needed to test and account for publication bias. Both of these aspects are critical in meta-analytic education research, the former providing another means of addressing comparability and boosting understanding, and the latter ensuring reliability and consistency of results. The final case study will serve as a modern example which accommodates for studies with differing sets of explanatory variables and addresses publication bias more directly, all within a specific application of multilevel modeling.

4.3 Meta-Regression

4.3.1 Method Description

Meta-regression is a special case of multilevel modeling, allowing for the estimation of several moderator variables effects simultaneously. The primary distinction between meta-regression and multilevel modeling is the former’s ignorance of the distinct sources of variance, a sacrifice typically preferred when the data only has two levels and/or only has aggregate-level data. Meta-regression is a simpler and thus more portable, often more calculable form of multilevel modeling which develops a predictive model of the variability of effect sizes. This flexibility leads to its frequent application in several research areas. Higgins, Lopez-Lopez, and Aloe (2021) argue that meta-regression should be conducted as ”part of a thorough systematic review to ensure that the studies involved are reliably identified and appraised”. This argument motivates the inclusion of meta-regression as a means of illuminating study comparability, and affirms its inclusion in the framework.

Meta-regression is a natural extension of linear regression, but with covariates at the study level rather than the individual level. As mentioned, it is a special case of multilevel modeling. It therefore ”borrows” the assumptions for linear regression and multilevel modeling. Each of the applicable assumptions listed in the previous section should be checked here as well. Naturally, τ^2 is assumed to be non-zero in meta-regression, and the specific meaning of the metric differs slightly in that it reflects the dispersion of true effects for studies with the same predicted value.

Meta-regression models can generally be notated as

$$\begin{aligned} y_i &\sim N(\theta_i, \sigma_i^2) \\ \theta_i &\sim N(x_i\beta, \tau^2) \end{aligned} \tag{7}$$

for studies $i = 1, \dots, K$, where y_i represents the estimated effect size of study i , normally distributed with its respective mean (true) effect size θ_i and variance σ_i^2 . θ_i is in turn assumed to be a realization of a distribution of all true effects. For J moderators, θ_i is parametrized by a linear combination of the vector of study-level covariates $x_i = (x_{i1}, x_{i2}, \dots, x_{iJ})$ and their respective coefficients $\beta = (\beta_1, \beta_2, \dots, \beta_J)$, and τ^2 . Note

that the coefficients are the same for each study, meaning the *moderating effects* are assumed to be *fixed*, not to be confused with the *random effect sizes*. The above is condensed to

$$y_i \sim N(x_i\beta, \sigma_i^2 + \tau^2) \quad (8)$$

giving the regression model $y_i = x_i\beta + v_i + \epsilon_i$, with $v_i \sim N(0, \tau^2)$ and $\epsilon_i \sim N(0, \sigma_i^2)$ which represent study-level variability and sampling variability, respectively. Thus, the estimated parameters of the model are $\Theta = (\beta, \tau^2, \sigma_i^2)$. An intercept coefficient β_0 may also be included in the model specifications and thus in Θ .

The computation of estimates for these parameters first involves estimating τ^2 , usually by DerSimonian-Laird / method of moments, maximum likelihood, restricted maximum likelihood (REML), or Empirical Bayes. As mentioned briefly in the multilevel modeling section, each method has strengths and weaknesses, so it is recommended to test as many as possible to check the robustness of results. When OLS is not easily applied, the fixed effect parameters β are estimated by conventional iterative methods such as the Newton-Raphson algorithm. σ_i^2 is typically estimated by REML or method of moments.

Once the parameters of the model have been estimated, significance tests can be applied to check whether the moderator is related to the effect sizes. As in sub-group analysis, these tests are often accompanied with the calculation of the proportion of variance explained by the moderator. Easily calculated from the fitted model is R^2 , the proportion of heterogeneity explained by the moderator variable(s), another effective indicator of the extent to which the model accounts for comparability concerns. These methods provide a natural approach to confirm moderators affecting results between studies. They allow for the estimation of the *strength* of effect of specific moderators on effect size variability, which can be contextualized in the comparability framework. The magnitude of parameter estimates for specific study-level covariates can be indicative of comparability issues in that a variable causing excessive discrepancies in effect sizes may imply fundamental differences between the studies. However, concerns may be reduced if significant predictors are determined. Since meta-regression models directly incorporate the effect of these moderators, they control for the discrepancies they cause and therefore model the data more accurately, strengthening the validity of conclusions.

Meta-regression is not a perfect approach. Its results are weakened in the presence of multicollinearity among predictors and has low statistical power which inflates significance when there are many moderators and/or a small number of studies. It is desirable to have a large number of studies relative to the number of moderators under inspection. As mentioned, REML is usually recommended in small-sample settings. Another shortcoming is the ecological fallacy, wherein relationships between study-level covariates and effect sizes may not be applicable to individuals. Methods exist to address these issues or assumption violations, such as generalized linear models, robust regression models, and spline regression, but they are out of the scope of this report.

4.3.2 Paper Introduction and Background

A pertinent example of meta-regression in educational research is Benos's and Zatou's *Education and Economic Growth: A Meta-Regression Analysis* from 2009. Human capital, or the set of knowledge, skills, and abilities embodied in individuals, is frequently acquired through education and training. The effect of human capital via education on economic growth is a highly debated topic and is challenging to measure for a number

of reasons. There exist many discrepancies in how growth and education are measured around the world, along with data quality issues, and the frequent disregard of vocational education (that is, examining only schooling). There are differing theories about the nature of economic growth via human capital, and moreover, measurements of education themselves are generally imperfect proxies. There is also variation in study characteristics, model specifications, data type, and publisher quality. The authors seek to reconcile these concerns by creatively and thoroughly applying meta-regression. Their work was further motivated by the detection of substantial publication bias toward positive results. The development of meta-regression models helps to explain how growth effect varies according to several factors, controlling for issues of comparability in the process.

4.3.3 Implementation and Results

Once again, the authors established rules for study inclusion. Although not as rigorous as the first case study, they ensured that the studies had some measure of growth as a dependent variable along with at least one education measure among the explanatory variables. The authors included three categories of studies: cross-section (growth differences between countries/regions), panel (cross-sectional and performance over time), and time-series (country-specific growth experiences). The following model was used:

$$\beta_j = \beta_0 + \sum_{k=1}^K \alpha_k Z_{jk} + \beta_1 se_j + u_j \quad (9)$$

β_j is the estimated education coefficient for the j -th study, β_0 is the true overall value of the education coefficient. Z_{jk} contains moderator variables that influence the magnitude of discrepancies and explain the variability in β_j . α_k is the coefficient for the respective moderator variable. Note that each study has the same covariate coefficients. se_j and u_j capture the standard error of the coefficient for study j and overall disturbance, respectively. The former is a means of addressing publication bias due to discrepancies in study size and thus effect magnitude. To compensate for widely varying sample sizes in education research, the following weighted least squares regression model is estimated:

$$t_j = \beta_1 + \sum \gamma_i K_{ij} + \beta_0/se_j + \sum \alpha_k Z_{jk}/se_j + v_j \quad (10)$$

t_j is the t -statistic that corresponds to the estimate of β_j . This specification gives more weight to studies with higher precision and less weight to studies with lower precision. An adjustment is made to β_1 in the form of $\sum \gamma_i K_{ij}$ which factors in differences between publication outlets to account for potential bias. K_{ij} contains only sample size as a moderator, while Z_{jk} contains all other moderators, each encoded as dummy variables. The inclusion of the following moderators allowed the authors to accommodate for different categories of studies, differences in measures of education, differing publication characteristics, etc. The authors emphasized the adherence to recommendations on sample size versus number of predictors in variable selection. Moderators tested by the authors are included below, omitting a few to maintain brevity.

- *Education variables*: indicate which of literacy rate, enrollment, schooling years (proportion of working-age population with each level of education), student-teacher ratios, educational expenditure, and scores were utilized as human capital proxies
- *Outcome measure*: indicate if GDP growth was used as a proxy for economic growth

- *Publication characteristics*: indicate where the studies were published and their ABS Economics journals classification grade, along with publication year
- *Estimation and data*: indicating where OLS methods were used, the usage of cross-sectional or panel data, and if the study came from specific popular data sets
- *Empirical specification*: indicating whether the study used logarithmic or linear model specification, and inclusion of explanatory variables (education, openness, political measures, government spending, population growth, inflation, health, investment, etc)

57 studies and 989 effect size estimates were utilized to fit the meta-regression models. The analysts began by fitting models with all variables in K and Z , then removing insignificant covariates one at a time and refitting the models. Models were estimated using four different techniques: cluster data analysis (CDA), REML, method of moments, and Empirical Bayes. Since the models employed a large number of variables, summary tables are omitted from this review. The authors reported the results of the models according to which publication indicator variable was used, which were then repeated after excluding the top 5% of extreme effect values. The main conclusions are as follows.

- *Education variables*
 - The proportion of working-age individuals with primary, secondary, and tertiary education was statistically insignificant in all models except CDA
 - Proxies based on enrollment rates and education expenditure increase the effect on economic growth by 0.01 and 0.02, respectively
 - Student-teacher ratios, literacy rates, schooling years, and test scores have statistically insignificant effects on the relationship between education and economic growth
- *Output measure*
 - No significant difference detected between GDP growth or GDP per capita
- *Publication characteristics*
 - Academic journals usually reported lower coefficients by about 0.005 compared to research published in working papers
 - Higher quality research outlets (higher ABS grades) published smaller education effects on economic growth
 - Publication year had an inconsistent effect
- *Empirical specification*
 - Inclusion of variables relating to enrollment, spending, political measures, initial output, and inflation tended to increase the impact of education on economic growth, even when corrected for publication bias
 - The use of openness, public spending, health indicators tended to lower the estimated impact of education on economic growth

These results were generally robust to varying model specifications and estimation methods. The authors conclude that none of the methods indicated a genuine education effect on growth after correction for publication bias. They identify varying data types, model specifications, and research outlet quality as the source of variation in empirical estimates. This is supported by the statistical significance of moderators, and the consistent production of R^2 statistics large by meta-regression standards ($R^2 = 0.16 - 0.78$).

4.3.4 Discussion

This is an excellent demonstration on the value of moderator analysis and maximizing statistical validity despite comparability concerns. Meta-regression, by nature, works to capture the sources of variability between studies. As such, its quality depends on the moderators used. If better predictors are used, a higher proportion of heterogeneity (and thus a greater extent of comparability) is accounted for. The authors obtained what they described as satisfactory R^2 values along with several statistically significant moderators, meaning their paper affirms the use of meta-regression as a method to address study incomparability. It could be argued that insignificant moderators reduce comparability concerns, because it signifies that a factor which *may* have caused discrepancies *did not*.

More specifically, the authors validated the comparison of the selected studies by encoding variables that accounted for discrepancies in education systems around the world in addition to study design differences. By comparing publication sources, the authors both ensured the robustness of results and estimated the effect of publication bias directly. Lastly, they demonstrate the value in testing many models across several computational techniques, augmenting the validity of their findings by showing robustness.

To critique the paper, it could be said that random effects for the covariates is desired. This would allow researchers to make inferences about the level of variability of moderating effects among studies. Additional subject-level covariates may also be considered. These extensions can be made with a multilevel modeling approach, which may come with additional computational complexity. It may also be argued that the models risk overfitting due to the number of parameters included. They note that the (albeit limited) non-robustness in results are due to econometric issues such as reverse causality, non-linearity, and parameter heterogeneity. In general, as is often the case in statistics, the results of this paper may still struggle in generalizing to the wider literature on the subject, despite its inclusion of a wide range of sources from various publications. It is infeasible to collect *all* studies from all time on the subject, while also controlling for publication bias and the factors which make the studies different. However, Benos and Zatou provide a quality case study on the value of meta-regression in educational research.

5 Summary and Closing Thoughts

The value of meta-analysis is reduced if it is not executed properly. An effective meta-analysis involves: 0) clear definitions of research objectives, 1) a detailed literature review that establishes study inclusion criteria, 2) computation of summary effects and heterogeneity statistics to detect between-study variability, 3) completion of at least one type of moderator analysis to determine sources of effect size variability, 4) an assessment of publication bias and outliers, and 5) the drawing of conclusions and reporting of results from all previous steps. This framework addresses meta-analytic criticisms, in particular the question of comparability, but it is not without limitations.

The proposed framework depends on the quality and quantity of data available. This is true for statistical methods in general and motivates the development of strict inclusion criteria to ensure relevance. *Inclusion criteria may be the most important factor in meta-analysis* because the sample must be generalizable to wider populations. If a study is very small or irrelevant, its inclusion in a meta-analysis may invalidate results, even if moderator analysis is applied to adjust for shortcomings. The meta-regression paper is an excellent example of reconciling studies which differ greatly in terms of sample size and measurement techniques, but the variability caused by these factors can only be controlled to a certain extent. A natural downside to rigorous study selection is that certain research areas may have a small number of studies to begin with, making it difficult to collect enough data. Effective meta-analysis requires a delicate balance, where researchers ideally include as many studies as possible, but the inclusion of more studies often requires the estimation of more moderator variables (see Benos and Zotou). This may then raise overfitting concerns and shift focus too strongly onto the moderator analysis instead of effect size estimation. It is easy to become sidetracked and spend too much time collecting and adjusting for studies, rather than analyzing results and implementing changes. Including too few studies makes generalizations harder to make. Study selection poses the greatest challenge in meta-analysis, but an effective sample is always worth the time and resource costs because it helps to validate statistical conclusions.

The moderator analysis methods covered are imperfect and come with assumptions that may not hold in all settings. Sub-group analysis is the most robust, as it simply involves the separation of studies according to some categorical factor. However, it is limited in that it does not exploit the data structure by proposing any formal model and may result in very small groups depending upon data availability in certain research areas. Multilevel modeling is more flexible in that it accommodates continuous and categorical moderators and can analyze multiple moderators simultaneously. Multilevel models are ideal for capturing the nested variance structure in meta-analytic samples, but the assumptions may be violated if publication bias significantly skews residuals or parameter estimates. Meta-regression is more case-specific than multilevel modeling, as its analysis is done on the aggregate level alone. It is thus useful for examining study-level moderators, but should be extended to multilevel modeling when dealing with complex, nested data structures. Meta-regression also inherits the potential assumption setbacks from multilevel methods. Additionally, combining several estimation techniques provides a strong analytical understanding and boosts reliability by testing result robustness.

Publication bias and outlier analysis are not discussed in great detail but are just as important a consideration as study comparability. The bias toward publishing significant results is common, especially in the social sciences. Significant bias can lead to violations of the assumptions within multilevel modeling and meta-regression and obfuscate conclusions in sub-group analysis. Researchers must also ensure that outlier results do not skew overall conclusions or cause spurious conclusions about moderators. As demonstrated by the case studies, moderator analysis is a useful tool for assessing both of these facets.

Ultimately, research quality is a function of the researcher themselves, owing to the "artistry" of statistical modeling. Statistical methods often require some creative balancing of strengths and weaknesses. Although perfect methods may not exist, there are techniques which address and reduce the effects of shortcomings. This was evidenced for meta-analysis in educational research by the selected case studies. In any case, education research must be invested consistently and continuously as a constantly evolving field.

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